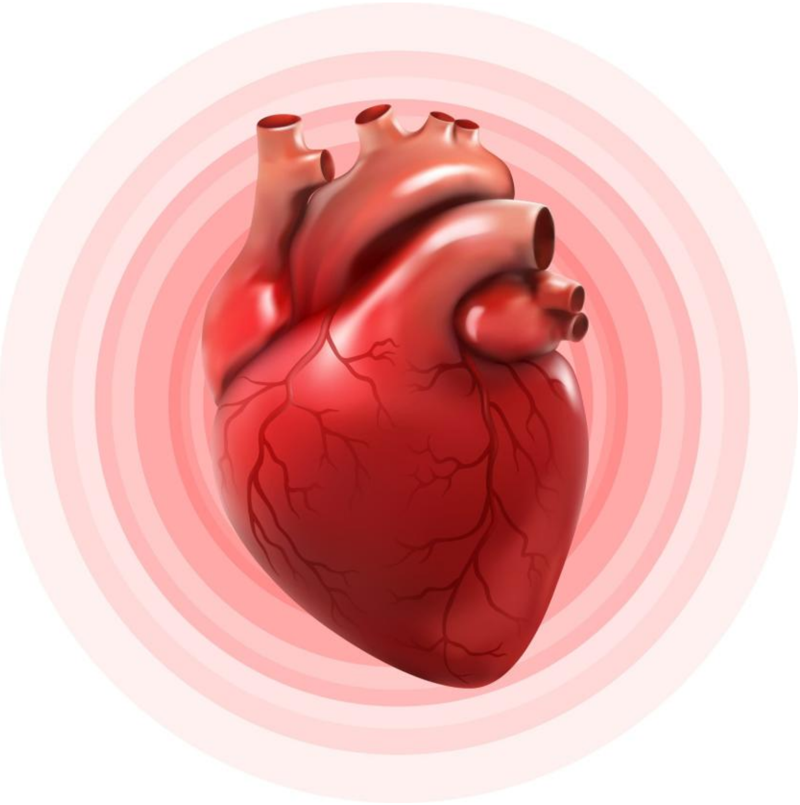

CORRECTING IRON DEFICIENCY IN HEART FAILURE

A Case Report

Importance of 1000 mg Ferric Carboxymaltose (FCM) in Heart Failure Patients with Newly Diagnosed Iron Deficiency
- Case Report

Case presentation:

- A 68-year-old male patient presented to the outpatient department (OPD) with a three-week history of worsening dyspnea, orthopnea, and ankle edema.
- The patient reported difficulty sleeping at night and a history of paroxysmal nocturnal dyspnea.
- Despite being on treatment for heart failure for the past five years, including guideline-directed medical therapy (GDMT) and lifestyle modifications such as water restriction, his symptoms had progressively worsened.

Medical History:

- Hypertension for 15 years.
- Diabetes mellitus for 5 years.
- Myocardial infarction 5 years ago.
- Heart failure with reduced ejection fraction (HFrEF) managed on maximal tolerated GDMT for the past 5 years.

Condition:

Coronary Artery Disease (CAD).

Intervention History:

The patient underwent a series of cardiovascular interventions to manage CAD and HFrEF:

- Multivessel PTCA (2000).
- PTCA (2016, 2020).
- CRT-D implantation (2020) with significant QRS narrowing achieved.

Comorbidities:

Type 2 Diabetes Mellitus, managed with oral hypoglycemic agents (fairly well-controlled).

Presenting Complaints:

- Progressive breathlessness and fatigue worsening over three years, with significant deterioration in the last month.
- Fatigue and reduced appetite.

Clinical Course:

- 2019: First admission for HFrEF.
- 2020: Second admission for HFrEF and CRT-D implantation.
- 2022: Two hospital admissions for HFrEF.
- Since 2021: On quadruple GDMT, including SGLT2 inhibitors and Ivabradine.
- February 2023: Received intravenous iron therapy.
- Emergency department visits requiring IV diuretics: Four episodes in the past year.

Clinical Examination:

- Pulse: 106/min (sinus rhythm)
- Blood Pressure: 96/58 mmHg
- Oxygen Saturation: 92% at rest
- Jugular Venous Pressure: Raised
- Peripheral Edema: Present
- Chest Examination: Bilateral fine crepitations at lung bases
- Cardiovascular System: Presence of S3 gallop

Laboratory Investigations:

- **Echocardiography (20th May 2023):**
  - Dilated left atrium and left ventricle
  - Moderate mitral regurgitation
  - Left Ventricular Ejection Fraction (LVEF): 22%
- **Biochemistry & Hematology:**
  - Hemoglobin: 11.2 g/dL
  - Serum iron: 12.41 µg/dL (reference: 59.00 - 158.00)
  - Ferritin: 18.30 ng/mL
  - Transferrin Saturation (TSAT): 12%
  - Serum creatinine: 1.9 mg/dL
  - Estimated Glomerular Filtration Rate (eGFR): 32 mL/min
  - Sodium: 124 mEq/L
  - Potassium: 5.3 mEq/L
- **NT-proBNP:** 3662 pg/mL
- **Procalcitonin:** 0.5 ng/mL
- **Liver Function Tests:** Unremarkable
- **ECG:** Narrow QRS complex, sinus rhythm

Current Medications:

- Carvedilol: 3.125 mg twice daily
- Ivabradine: 5 mg twice daily
- Sacubitril/Valsartan: 50 mg twice daily
- Empagliflozin: 10 mg once daily
- Torasemide: 20 mg twice daily
- Eplerenone: 25 mg (half tablet post-lunch)
- Amiodarone: 100 mg post-lunch
- Trimetazidine: 80 mg after breakfast

Diagnosis:

- Coronary Artery Disease (CAD)
- Severe Left Ventricular Dysfunction
- Iron Deficiency

Management Plan:

- **Acute Management:**
  - Administer intravenous diuretics in the emergency department to manage fluid overload.
  - Treat iron deficiency based on iron studies.
- **Intervention:**
  - Patient received 1000 mg Ferric Carboxymaltose (FCM) intravenously over 20 minutes in 100 mL normal saline.
- **Dietary Advice:**
  - Strict diet modification to include iron-rich foods.

Follow-Up and Outcomes:

- The patient remained stable for one hour post-treatment and was discharged.
- One month later:
  - Patient was asymptomatic.
  - Could walk 30 minutes at a moderate pace without experiencing breathlessness.
  - Serum ferritin: 72 ng/mL
  - Transferrin Saturation (TSAT): 32%
- No serious adverse events (SAEs) or severe allergic reactions related to the treatment were reported.

Discussion:

Iron deficiency is a common comorbidity in heart failure with reduced ejection fraction, contributing to worsening symptoms, reduced exercise capacity, and increased hospitalizations.[1] In this case, a 68-year-old male with chronic heart failure and documented iron deficiency showed significant improvements following intravenous ferric carboxymaltose (FCM) therapy. The treatment effectively corrected iron parameters, as demonstrated by improved serum ferritin and transferrin saturation levels, while also enhancing the patient's functional capacity, enabling moderate physical activity without dyspnea.

The safety and tolerability of FCM in this case were evident, with no adverse events or hospitalizations reported. These findings underscore the effectiveness of intravenous iron therapy in improving clinical stability and quality of life. Routine screening and timely correction of iron deficiency are pivotal for alleviating symptoms and preventing hospitalizations in patients with advanced heart failure.[1,2]

CONCLUSION

This case illustrates the significant benefits of intravenous ferric carboxymaltose in managing iron deficiency in a patient with chronic heart failure and reduced ejection fraction. The therapy led to improved functional capacity, correction of iron parameters, and clinical stabilization without adverse effects.

Early identification and management of iron deficiency in heart failure patients are essential to reducing symptoms, minimizing hospitalizations, and enhancing overall outcomes. This case further supports evidence-based iron supplementation as a vital component of heart failure management.[1,2]

CAD, Coronary Artery Disease; CRT-D, Cardiac Resynchronization Therapy with Defibrillator; FCM, Ferric Carboxymaltose; GDMT, Guideline-Directed Medical Therapy; HFrEF, Heart Failure with Reduced Ejection Fraction; IV, Intravenous; LVEF, Left Ventricular Ejection Fraction; OPD, Outpatient Department; PTCA, Percutaneous Transluminal Coronary Angioplasty; QRS, QRS Complex (on an electrocardiogram); SGLT2 Inhibitors, Sodium-Glucose Co-Transporter 2 Inhibitors.

References:

1. Ponikowski P, Mentz RJ, et al. Efficacy of ferric carboxymaltose in heart failure with iron deficiency: an individual patient data meta-analysis. Eur Heart J. 2023 Dec 21;44(48):5077-5091. doi: 10.1093/eurheartj/ehad586. PMID: 37632415; PMCID: PMC10733736.

2. Bagirath R, Raghuraman R, et al. Prospective observational study to assess effectiveness and safety of intravenous ferric carboxymaltose 1000 mg in patients of chronic heart failure with iron deficiency - PRIDE-HF study. Cardiol Cardiovasc Med. 2023;7:349-54.